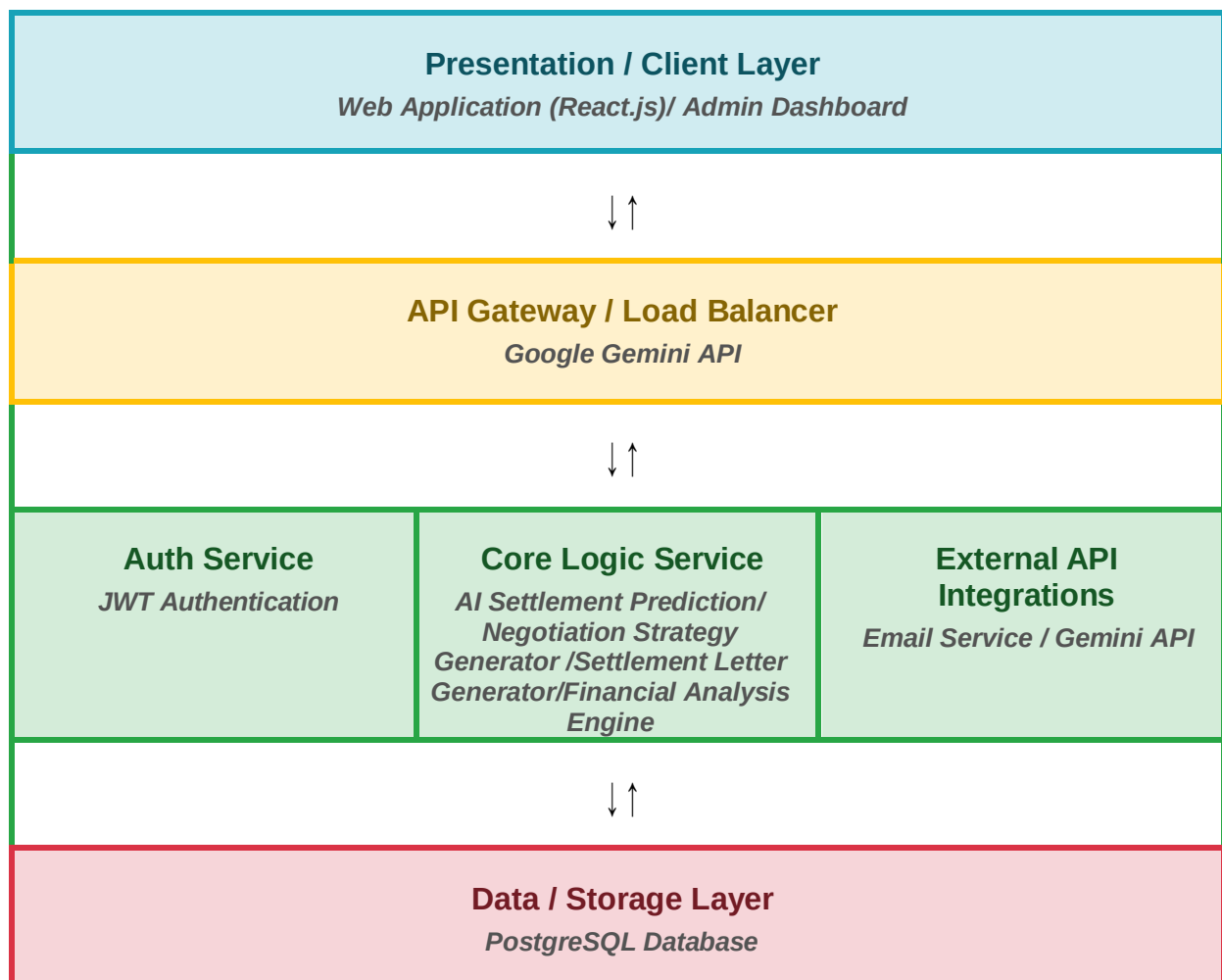


Solution Architecture

Date	15 JULY
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for borrowers and administrators to register, manage loans, view financial profiles, generate settlement recommendations, and access AI-powered financial recovery services.	React.js, HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Acts as the central entry point for all client requests, routes requests to backend services, manages authentication, validates APIs, and ensures secure communication between the frontend and backend.	Node.js, Express.js, REST API, JWT Authentication
Microservices	Implements the core business logic including user management, loan management, financial profile analysis, AI-powered settlement prediction, negotiation strategy generation, settlement letter generation, and history tracking.	Python, Flask, Scikit-learn, Pandas, NumPy, OpenAI API (or Gemini API)
Database	Stores borrower information, loan details, financial profiles, settlement records, AI interaction history, and application data while maintaining data integrity and supporting efficient retrieval and analytics.	PostgreSQL (or MySQL), SQLAlchemy ORM